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AI based intuitionistic dense fuzzy entropy decision- making system and its application in firefighting robot selection

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ABSTRACT

This study explores fuzzy entropy in the context of Intuitionistic Dense Fuzzy Sets (IDFS), considering the convergence of a sequence $x_n \in DFS(X)$ to $x \in DFS(X)$ as $n \to \infty$. Three novel formulations of Intuitionistic Dense Fuzzy Entropy (IDFE) are established using max–min operations, exponential and logarithmic functions. Based on performance evaluation, the most effective IDFE formulation is selected to handle Multi-Criteria Decision-Making (MCDM) problems. To enhance decision-making efficiency, an integrated AI based approach is proposed by developing a Multi-Objective Intuitionistic Dense Fuzzy Breadth-First Search (MO-IDF-BFS) algorithm. This algorithm extends the traditional BFS framework by incorporating multi-objective optimization principles and embedding it within the IDFE and weighted sum product (WISP)-based decision structure. The applicability of the proposed framework is established for a firefighting robot selection problem. The proposed MO-IDF-BFS algorithm determines an operational path that optimally balances travel distance and risk exposure, while the ranking of alternative robots is carried out using the integrated IDFE–WISP model. A comparative evaluation with the existing fuzzy MCDM models such as ARAS, WASPAS, COPRAS, WAS and WAP is carried out to validate the superior performance of the proposed system. Additionally, a machine learning–based validation using random forest regression and Leave-One-Out Cross-Validation confirms the reliability and internal consistency of the proposed rankings with a Spearman correlation coefficient of $\rho = 0.94$ between predicted and actual ranks. Further, sensitivity analysis and Monte-Carlo experiment analysis confirm the robustness, stability and interpretability of the proposed model. The results indicate that the Thermite RS3 robot is the most suitable choice for intelligent firefighting systems.

1. Introduction

Fire is a rapid chemical process that occurs when a material undergoes a process called combustion, which involves a chemical reaction between a fuel source, oxygen, heat and a chemical chain reaction. These four essential components of fire are often referred to as the “fire tetrahedron”.

A fire can be extinguished by removing any one of the four components of the fire tetrahedron. Utilizing water is one way to put out a fire. Water can be obtained through pressurized fire hydrants, pumped from rivers or lakes, transported by tanker trucks or dropped from aircraft tankers during the control of fires. A firefighter battles fires in order to save lives, property and the environment from being destroyed [1]. Firefighting is the act of extinguishing fires which is a highly technical skill that requires professionals who have spent years training in both general firefighting techniques and specialized areas of expertise. The hazardous atmosphere created by combusting items is one of the main threats involved in firefighting operations.

The amount of water required to put out the fire can be evaluated by calculating its fireload using the formula given in Eq. (1). Fireload refers to the maximum heat that would be released if all combustible materials within a designated fire area were to burn. Depending on the current ventilation conditions, the fireload within a building has a considerable impact on the fire’s severity, velocity of spread and longevity.

$$\text{Fireload} = \frac{\text{Weight of the material in kg (mass)} \times \text{Calorific value of the material in Joule}}{\text{Floor area in sq.meter}} \tag{1}$$

Approximately 6% of all recorded deaths from both human and natural causes are caused by fire. Fire catastrophes that traumatised humans include “The triangle shirtwaist fire”, New York City, USA, in the year

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1911, which caused about 146 deaths of which 123 were women and girls and 23 were men. Haryana school blaze, India, in the year 1995, caused the death of 500 people, including 258 children. Hospital fire, China, in the year 2023, where 21 were dead and 71 patients were evacuated. Recently, in India, Bhalswa landfill fire accident, New Delhi, in the year 2022, where 36 acres of garbage dumping which weighs around 80 lakhs tonnes faced a fire accident. This fire did not cause any loss of life but the smoke caused by the fire damaged lives more internally by affecting peoples respiratory systems. Peru gold mine fire, in may, 2023 caused the loss of 27.

Sometimes, it turns into a loss of life for firefighters, when working in these kinds of hazardous fire situations. Due to fire accidents, firefighters consistently face life-threatening situations. Thus, utilizing robots to assist firefighters helps them to put the fire out from a safer distance. Though robots save lives, they may not replace human firefighters entirely, but they can complement human capabilities in various ways. The selection of robots to help humans in firefighting is going to be a difficult task as humans bring a combination of physical abilities, cognitive skills and emotional intelligence. Therefore, employing robots for human protection necessitates meeting specific criteria, such as selecting the most efficient path and prioritizing the most suitable robot. Swethaa and Felix [2] stated that the usage of robots in dangerous fields helps in reducing the death rate of humans.

Selecting and prioritizing the most suitable robot is a difficult task as the data include fuzzy information that is often vague and imprecise. Dealing with imprecise information is challenging as uncertainty is involved in every real world system. Thus, the theory of fuzzy sets was introduced by Zadeh [3], in 1965 and it is capable of addressing uncertainty and vagueness. Since its inception, this theory has evolved into more advanced forms like vague sets [4], intuitionistic fuzzy sets [5], interval-valued intuitionistic fuzzy sets [6], Pythagorean fuzzy sets [7], dense fuzzy sets [8], intuitionistic dense fuzzy sets [2], bipolar complex fuzzy soft sets [9] and many more. These advanced theories are capable of handling more complicated problems and are commonly applied in fields such as image processing, supply chain management, risk analysis, medical diagnosis, and waste management.

Dense fuzzy set [8], where the uncertainty fades with respect to time is one of the specialized sets in handling fuzziness. However, the combination of two kinds of sets works better than that of normal fuzzy extensions. Thus, the intuitionistic dense fuzzy sets (IDFS) are considered to be more adequate to solve real time problems and applications. IDFS has elements that involve both membership and non membership. Thus, robot selection is performed in this context.

1.1. Aim and benefit of IDFS

When an element x of a set exhibits vagueness around itself, it directly relates to the closure or neighborhood set, which is dense. At this point, the fuzziness is reduced to x with respect to the decision maker's improvement through their learning experience, observation frequency, turnover or processing time. Thus, this process results in the development of DFS. Building on this notion, De and Beg [8] in 2016 introduced the triangular dense fuzzy set, where the elements are a converging sequence that converges to an element or an interval within the set when $n \to \infty$. To include the elements' non-membership with its hesitancy, the idea of IDFS was introduced by Swethaa and Felix [2] which helps in simplifying the process of defuzzification and ranking for intuitionistic dense fuzzy numbers (IDFN). The merits of employing IDFS are outlined below.

(i) IDFS aids by valuing the experience of the experts and prioritizing their opinions by increasing the value of n.
(ii) In IDFS, the vagueness decreases and eventually converges to an element or an interval within the set as the value of n increases.
(iii) Expert perspectives are taken into account for both membership and non-membership based on their experience.

1.2. Entropy for IDFS

As many real-world problems involve fuzziness, the measurement of fuzziness is also important so that the weightage can be assigned according to the fuzziness present in the element. Hence, this paved the way for the concept of entropy. In 1947, Shannon [10] introduced the idea of entropy, which calculates the level of disorder in a system. It also helps in assessing the degree of uncertainty in a system. In fuzzy, entropy is characterised as the measure of fuzziness within fuzzy set theory. Entropy for fuzzy sets was first introduced by De Luca and Termini in 1972 [11]. Burillo and Bustince [12] extended the concept of entropy for intuitionistic fuzzy sets. In 2012, Li et al. [13] and Deng et al. [14] extensively discussed the methodologies for constructing similarity-based entropy measures. Alreshidi et al. [15] expanded the entropy measure for circular intuitionistic fuzzy sets by incorporating the radius parameter. In the same year, Patel [16] introduced an entropy measure by utilizing the newly proposed similarity measure. These entropy measures have proven it's use across various fields. One of the major applications is that they are used in assigning weights for the criteria in multi-criteria decision making (MCDM) models. For instance, Szmidt [17] proposed a non-probabilistic entropy measure for IFS. Szmidt extended their previous work on entropy measures for IFS by exploring the differences using normalized Euclidean and Hamming distances [18].

Recent studies on the problem of robot selection using the MCDM model for the ITpDFN framework were carried out by Swethaa et al.[2, 24]. Uncertainty handling in prioritization methods is inherently challenging due to the presence of multiple, often conflicting criteria in the decision-making process. Zarei et al. [51] proposed a methodological framework for integrating expert knowledge into uncertainty analysis within sociotechnical systems. Yazdi et al. [52] highlighted the important role of uncertainty modeling in digitalized process safety by classifying sources of uncertainty. Therefore, by addressing more uncertainties, this research uses IDFS to deal with imprecise problems. Hence, the main objective of the paper is to combine entropy with the MCDM model under the IDF field to select the best firefighting robots. The entropy metric was used by Ali et al. [19] to rank IFSs and by Khan et al. [20] to rank qROFSs. Moreover, entropy methods were incorporated into MCDM models to enhance outcomes [22,23]. Mishra and Rani [21] introduced an ARAS method based on entropy measures for selecting sustainable recycling partners.

1.3. Review on MCDM techniques

MCDM is a widely used decision-making approach that incorporates both quantitative and qualitative aspects in a variety of domains and applications [25–32]. For instance, Vlsekriterijumska optimizacija kompromisno resenje (VIKOR), Technique for Order Performance by Similarity to Ideal Solution (TOPSIS), Elimination and Choice Translating Reality (ELECTRE), Preference Ranking Organization Method for Enrichment Evaluations (PROMETHEE) are a few widely used MCDM models. Every technique has benefits and drawbacks, and the optimal approach depends on the nature of the problem, the data available and the decision-makers' opinions. It's worth noting that the field of MCDM is constantly evolving, with new models and variations being introduced on a regular basis. In recent decades, MCDM techniques like the Complex Proportional Assessment (COPRAS), Simple Weighted Sum Product Method (WISP), Weighted Aggregated Sum Product Assessment (WASPAS) and Simple Additive Weighting Method (SAW) have emerged and gained popularity. Obtaining WISP which was first introduced by Stanujkic [33] in 2021. Yin et al. [53] proposed a hybrid navigation model to enhance safety and obstacle avoidance in dynamic environments. Shen [54] optimized a fuzzy neural network controller using a hybrid Particle Swarm Optimization–Backpropagation algorithm to improve stability, tracking accuracy and robustness of robotic manipulator systems. Ho et al. [55] proposed a reinforcement learning approach for path planning of dynamic automated guided vehicles. Ding et al. [56] developed a situation assessment algorithm for

accurate fire detection and analysis in firefighting robots. Lin et al. proposed novel directional correlation coefficients for Pythagorean fuzzy sets and the proposed coefficients were applied to disease diagnosis and cluster analysis to demonstrate their practical effectiveness [37]. Huang et al. proposed novel Dice and Jaccard distance measures and a new score function to develop a novel Pythagorean fuzzy MULTIMOORA method. The proposed method was applied for evaluating solid-state disk productions and energy projects to verify the proposed method [38]. To aggregate the linguistic q-rung orthopair fuzzy numbers, Lin et al. developed linguistic q-rung orthopair fuzzy interactional PGHM (LqROFIPGHM) operator and linguistic q-rung orthopair fuzzy interactional weighted PGHM (LqROFIWPGHM) operator [39]. The most recently introduced MCDM method, has drawn interest in decision-making because of its ability to handle challenging decision problems and offer insightful information for choosing alternatives. Hence, a novel IDFE-WISP MCDM model is designed to prioritize firefighting robots.

1.4. Review on AI techniques

Intelligent systems play a pivotal role in preventing, managing and responding to mine accidents and emergencies. Enhancing intelligent systems contributes to a safer mining process and reduces the frequency and severity of mine accidents [40]. Yagüe et al. integrated expert knowledge with Particle Swarm Optimization and Pattern Search to enhance robotic trajectory performance while reducing computational burden [47]. To determine the optimal or close-to-optimal parameters for high-quality walking of the soft robot, Tan et al. applied stochastic black-box optimization to solve the proposed multi-fidelity Bayesian optimization [41]. García and Duarte identified key design parameters for effective obstacle navigation in the development of mobile robots [44]. Paul and Machavaram developed a comprehensive design and optimization framework for a 6-DOF robotic manipulator intended for vision-guided capsicum harvesting in greenhouse environments. To enhance positioning accuracy, a hybrid AI-based optimization method was formulated by combining neural networks, fuzzy logic, and genetic algorithms [42]. Hemalatha et al. introduced an integrated intelligent system with a robotic arm for object detection to achieve autonomous cleaning and object interaction in dynamic home environments [43]. In order to quantify cost and energy savings, Klar et al. designed a hybrid optimization model for robot–van parcel delivery [45]. A structured review of snake robot design evolution, structural configurations, sensing and control was conducted by Moosavi et al. The review also specified application domains such as search and rescue, industrial inspection and environmental exploration [46]. Luer et al. presented a comprehensive review of magnetically actuated micro- and nano robotic systems integrated with AI-based optimization and navigation strategies [48]. Yang et al. introduced a real-time vision-based framework that combines YOLOv5s for high-accuracy flame detection with stereo vision for 3D position estimation, specifically optimized for embedded deployment [49]. Deng et al. developed a CCTV informed autonomous navigation system that enables firefighting robots to reach target flames through the safest and shortest path [50]. Although AI techniques are applied in various domains, the integration of uncertainty modeling using IDFS with AI-based optimization and decision-making approaches remains limited. Therefore, the present study incorporates IDFS-based uncertainty modeling within an AI-supported decision framework, enabling more reliable and robust decision-making.

1.5. Motivation

Existing literature indicates a gap in research concerning the concept of IDFS that exclusively focuses on the belongingness and non-belongingness functions. Consequently, it is evident that there is a lack of studies centered around the concept of IDFE and AI based IDF decision-making models. Addressing this research gap is a primary objective of the present study, which aims to answer the following research questions, listed below.

(i) How do converging functions operate on IDFS, incorporating both the belongingness and non-belongingness values of the elements?
(ii) Is it possible to formulate an IDF entropy that satisfies all four axioms, addressing drawbacks such as counter-intuitive and unreasonable results, the zero divisor problem, violations of entropy axioms and violations of the crisp property?
(iii) How can BFS be executed under an IDF environment when there are multiple source and sink nodes?
(iv) How can two values which determine the two characteristics of a path be aggregated to calculate the overall value of the path?
(v) How can AI based IDF decision making models be developed?

Therefore, this research gap and the arising questions serve as the motivation for conducting the present study, as outlined below.

(i) The literature reveals that no study has provided a solution for measuring fuzziness under IDFS. Hence, this serves as a primary motivation to introduce a better performing IDFE.
(ii) The determination of an IDFE that can handle all the drawbacks is still in progress.
(iii) This study is also motivated by introducing AI based decision making system by integrating the IDFE and MO-IDF-BFS models to find the weights of the criteria.
(iv) The challenge of finding the shortest path in any network where the path values are derived by integrating both distance and risk levels is determined.
(v) The lack of integration AI based optimization techniques with the recent MCDM models in the existing literature motivates the need to find the shortest path for any network to operate and to prioritize the best alternatives. This helps in enhancing decision-making effectiveness in various domains.
(vi) There is a lack of application in utilizing firefighting robots to replace humans. This can be addressed through MCDM approaches, aiming to minimize the human death rate in firefighting operations.

1.6. Contribution

MCDM models are very powerful tools for making decisions when considering multiple criteria. This study has made a substantial contribution and introduced major innovations to the area by analyzing gaps in the existing literature.

The main contributions of the paper are summarized below:

(i) Novel entropy formulations are derived under IDFS. The proposed entropy functions are designed to better represent uncertainty when n grows larger.
(ii) To determine optimal paths by jointly considering distance and risk factors, the MO-IDF-BFS algorithm is proposed.
(iii) An integrated decision-making framework which combines the proposed entropy measure with the WISP model is developed for the firefighting robot selection problem. A linguistic scale for IDFS is derived to capture decision-makers' experiential judgments. To check the stability and consistency of the obtained results, comparative analysis, sensitivity and ML validation are conducted.

The paper is organized as follows: Section 2 delves into the fundamental IDFS concepts. Section 3 introduces the novel entropy measures for IDFS with the axioms, theorems, and examples. Section 4 comprises the MO-IDF-BFS algorithm to find the shortest path and the IDFE-WISP algorithm to prioritize the alternatives. Additionally, the comparative analysis, result and discussion of the illustrative example are presented. Lastly, Section 5 provides a conclusion of the study along with recommendations for future research.

2. Study context

In this section, fundamental definitions of IDFS are explored. Basic operations for IDFS, along with concepts of score and accuracy functions are elucidated.

Definition 2.1 ([34]). A fuzzy set $\tilde{A} = \{\ldots, x_{-2}f_n, x_{-1}f_n, x_0, x_1g_n, x_2g_n, \ldots\}$ is considered where the elements are converging sequences that converge to x_0 as $n \to \infty$. If all the elements of $\tilde{A}$ converge to an element x_0 as n increases then the considered set is known as a dense fuzzy set (DFS).

Definition 2.2. An intuitionistic dense fuzzy set (IDFS) $\tilde{A}$ is defined as $\tilde{A} = \{(x, f_n * \mu_{\tilde{A}}(x,n), g_n * \gamma_{\tilde{A}}(x,n)) \mid x \in X\}$, where $\mu_{\tilde{A}}(x,n) : X \to [0,1]$, $\gamma_{\tilde{A}}(x,n) : X \to [0,1]$, f_n and g_n are functions that converge to particular $\mu_{\tilde{A}}(x,n)$ and $\gamma_{\tilde{A}}(x,n)$ values of x as $n \to \infty$, $\mu_{\tilde{A}}(x,n), \gamma_{\tilde{A}}(x,n) \in [0,1]$ and $0 \le f_n * \mu_{\tilde{A}}(x,n) + g_n * \gamma_{\tilde{A}}(x,n) \le 1$.

For simplicity, this study incorporates IDFS, which can be expressed as $\tilde{A} = \{(f_n * \mu_{\tilde{A}}), (g_n * \gamma_{\tilde{A}})\}$ where f_n and g_n are the converging functions, $\mu_{\tilde{A}}, \gamma_{\tilde{A}} \in [0,1]$ and $0 \le \mu_{\tilde{A}} + \gamma_{\tilde{A}} \le 1$ and the hesitancy part $\pi_{\tilde{A}}$ can be expressed as $\pi_{\tilde{A}} = 1 - \mu_{\tilde{A}} - \gamma_{\tilde{A}}$

Definition 2.3. An $\tilde{A}_k = \{(f_n * \mu_{\tilde{A}}), (g_n * \gamma_{\tilde{A}})\}$, $k = 1, 2, \ldots, n$ is an IDFS. Then, the score $S(\tilde{A}_k)$ and accuracy $H(\tilde{A}_k)$ values are given in Eq. (2)

$$S(\tilde{A}_k) = (f_n * \mu_{\tilde{A}}) - (g_n * \gamma_{\tilde{A}})\ ;\ H(\tilde{A}_k) = (f_n * \mu_{\tilde{A}}) + (g_n * \gamma_{\tilde{A}}) \tag{2}$$

where $S(\tilde{A}_k) \in [-1,1]$ and $H(\tilde{A}_k) \in [0,1]$. Subsequently, Xu [35] modified this definition which is given in Eq. (3)

$$S^*(a_k) = \frac{1}{2}(S(\tilde{A}_k) + 1)\ ;\ H^*(a_k) = 1 - H(\tilde{A}_k) \tag{3}$$

Definition 2.4. Let $\tilde{A} = \{(f_n * \mu_{\tilde{A}}), (g_n * \gamma_{\tilde{A}})\}$ and $\tilde{B} = \{(f_n * \mu_{\tilde{B}}), (g_n * \gamma_{\tilde{B}})\}$ be two IDFS on X. Below are several fundamental operations that can be applied to IDFSs $\tilde{A}$ and $\tilde{B}$.

1. $\tilde{A} \subseteq \tilde{B}$ iff $(f_n * \mu_{\tilde{A}}) \le (f_n * \mu_{\tilde{B}})$, $(g_n * \gamma_{\tilde{A}}) \le (g_n * \gamma_{\tilde{B}})$
2. $\tilde{A} = \tilde{B}$ iff $(f_n * \mu_{\tilde{A}}) = (f_n * \mu_{\tilde{B}})$, $(g_n * \gamma_{\tilde{A}}) = (g_n * \gamma_{\tilde{B}})$
3. $\tilde{A}^c = \{(g_n * \gamma_{\tilde{A}}), (f_n * \mu_{\tilde{A}})\}$, where $\tilde{A}^c$ is the complement of $\tilde{A}$.
4. $\tilde{A} \cup \tilde{B} = \{\max(f_n * \mu_{\tilde{A}}, f_n * \mu_{\tilde{B}}), \min(g_n * \gamma_{\tilde{A}}, g_n * \gamma_{\tilde{B}})\}$
5. $\tilde{A} \cap \tilde{B} = \{\min(f_n * \mu_{\tilde{A}}, f_n * \mu_{\tilde{B}}), \max(g_n * \gamma_{\tilde{A}}, g_n * \gamma_{\tilde{B}})\}$

3. Entropy measures for IDFS

This section delves into the concept of entropy, which plays a crucial role in quantifying the degree of fuzziness within the realm of IDFS.

Definition 3.1. A mapping $E : IDFS(X) \to \mathbb{R}^+$ is said to be an entropy measure on IDFS(X) if E satisfies the following properties, where $\tilde{A} = \{(x, \mu(x_i,n), \gamma(x_i,n)) \mid x \in IDFS(X)\}$

(i) $E(\tilde{A}) = 0$ iff $\tilde{A}$ is a crisp set
(ii) $E(\tilde{A}) = 1$ if $f_n * \mu_{\tilde{A}}(x_i,n) = g_n * \gamma_{\tilde{A}}(x_i,n)$, $\forall\ \tilde{A} \in IDFS(X)$
(iii) $E(\tilde{A}) = E(\tilde{A}^c)$, $\forall\ x_i \in \tilde{A}$
(iv) $E(\tilde{A}) \le E(\tilde{B})$ iff $f_n * \mu_{\tilde{A}}(x_i,n) \le f_n * \mu_{\tilde{B}}(x_i,n)$ and $g_n * \gamma_{\tilde{A}}(x_i,n) \ge g_n * \gamma_{\tilde{B}}(x_i,n)$, for $f_n * \mu_{\tilde{B}}(x_i,n) \le g_n * \gamma_{\tilde{B}}(x_i,n)$ or if $f_n * \mu_{\tilde{A}}(x_i,n) \ge f_n * \mu_{\tilde{B}}(x_i,n)$ and $g_n * \gamma_{\tilde{A}}(x_i,n) \le g_n * \gamma_{\tilde{B}}(x_i,n)$, for $f_n * \mu_{\tilde{B}}(x_i,n) \ge g_n * \gamma_{\tilde{B}}(x_i,n)$

3.1. Entropy measure based on max − min function

Definition 3.2. Let $\tilde{A} = \{((x_i,n), \mu(x_i,n), \gamma(x_i,n)) \mid x \in X\}$ be an IDFS. Then, the entropy of $\tilde{A}$ on the max − min function is formulated by the following Eq. (4)

$$E_1(\tilde{A}) = \frac{1}{mN}\sum_{i=1}^{m}\sum_{n=1}^{N} \times \left(\frac{2 - |f_n * \mu_{\tilde{A}}(x_i,n) - g_n * \gamma_{\tilde{A}}(x_i,n)| - \max_i |f_n * \mu_{\tilde{A}}(x_i,n) - g_n * \gamma_{\tilde{A}}(x_i,n)|}{2 + \min_i |f_n * \mu_{\tilde{A}}(x_i,n) - g_n * \gamma_{\tilde{A}}(x_i,n)| - \max_i |f_n * \mu_{\tilde{A}}(x_i,n) - g_n * \gamma_{\tilde{A}}(x_i,n)|}\right) \tag{4}$$

Theorem 3.3. *The $E_1(\tilde{A})$ defined in Eq. (4) is an IDF entropy if it satisfies the Definition 3.1 .*

Proof. If a mapping satisfies the axiomatic properties listed in Definition 3.1, it can be classified as an entropy measure. Therefore, the axioms for Eq. (4) are validated here.

A1: If $E_1(\tilde{A}) = 0$, then $\tilde{A}$ is a crisp set, $\forall\ \tilde{A} \in IDFS(X)$.

Assuming, $\mu(x_i,n) = 1$, $\gamma(x_i,n) = 0$ or $\mu(x_i,n) = 0$, $\gamma(x_i,n) = 1$, then from Eq. (4)

$$E_1(\tilde{A}) = \frac{1}{mN}\sum_{i=1}^{m}\sum_{n=1}^{N} \times \left(\frac{2 - |f_n * \mu_{\tilde{A}}(x_i,n) - g_n * \gamma_{\tilde{A}}(x_i,n)| - \max_i |f_n * \mu_{\tilde{A}}(x_i,n) - g_n * \gamma_{\tilde{A}}(x_i,n)|}{2 + \min_i |f_n * \mu_{\tilde{A}}(x_i,n) - g_n * \gamma_{\tilde{A}}(x_i,n)| - \max_i |f_n * \mu_{\tilde{A}}(x_i,n) - g_n * \gamma_{\tilde{A}}(x_i,n)|}\right)$$

Consider, $|f_n * \mu_{\tilde{A}}(x_i,n) - g_n * \gamma_{\tilde{A}}(x_i,n)| = \sigma(x_i,n)$

Then, $\forall x_i \in X$ and when $N \to \infty$ the result is trivial. Thus, $E_1(\tilde{A}) = 0$ is proved.

A2: If $E_1(\tilde{A}) = 1$,

Assuming, $f_n \times \mu_{\tilde{A}}(x_i,n) = g_n \times \gamma_{\tilde{A}}(x_i,n)$, $|f_n * \mu_{\tilde{A}}(x_i,n) - g_n * \gamma_{\tilde{A}}(x_i,n)| = \sigma(x_i,n)$

$$E_1(\tilde{A}) = \frac{2 - |f_n * \mu_{\tilde{A}}(x_i,n) - g_n * \gamma_{\tilde{A}}(x_i,n)| - |f_n * \mu_{\tilde{A}}(x_i,n) - g_n * \gamma_{\tilde{A}}(x_i,n)|}{2 + |f_n * \mu_{\tilde{A}}(x_i,n) - g_n * \gamma_{\tilde{A}}(x_i,n)| - |f_n * \mu_{\tilde{A}}(x_i,n) - g_n * \gamma_{\tilde{A}}(x_i,n)|} = 1\ \forall x_i \in X$$

and when $N \to \infty$ the result is trivial.

A3: If $E_1(\tilde{A}) = E_1(\tilde{A}^c)$

We know that, $\tilde{A}^c = \{(x_i, \gamma(x_i,n), \mu(x_i,n)) | x_i \in X\}$.

i.e., $\mu_{\tilde{A}}(x_i,n) = \mu_{\tilde{A}^c}(x_i,n)$ and $\gamma_{\tilde{A}}(x_i,n) = \gamma_{\tilde{A}^c}(x_i,n)$

Thus, $\sigma_{\tilde{A}}(x_i,n) = \sigma_{\tilde{A}^c}(x_i,n)$.

When $N \to \infty$, the result is straightforward.

A4: If $\tilde{A}$ and $\tilde{B}$ are two IDFS, $E(\tilde{A}) \le E(\tilde{B})$, the conditions $\mu_{\tilde{A}}(x_i,n) + \gamma_{\tilde{A}}(x_i,n) \ge \mu_{\tilde{B}}(x_i,n) + \gamma_{\tilde{B}}(x_i,n)$

Hence, $E(\tilde{A}) \le E(\tilde{B})$. □

3.2. Entropy measure based on exponential function

Definition 3.4. If $A = \{((x_i,n), \mu(x_i,n), \gamma(x_i,n))\}$ is an IDFS, then the entropy measure on the exponential function $(E2)$ is defined by the following Eq. (5) .

$$E_3(\tilde{A}) = \frac{1}{mN}\sum_{i=1}^{m}\sum_{n=1}^{N}\left[\frac{e^{-|\mu_{\tilde{A}}(x_i,n) - \gamma_{\tilde{A}}(x_i,n)|} - e^{-1}}{1 - e^{-1}} + \left(1 - |\mu_{\tilde{A}}(x_i,n) - \gamma_{\tilde{A}}(x_i,n)|\right)\right] \tag{5}$$

Theorem 3.5. *The $E_2(\tilde{A})$ defined in Eq. (5) is an IDF entropy if it satisfies the Definition 3.1 .*

Proof. If a mapping satisfies the axiomatic properties listed in Definition 3.1, it can be classified as an entropy measure. Therefore, the axioms for Eq. (5) are validated here.

A1: If $E_3(\tilde{A}) = 0$, then we have,

$$\mu(x_i,n) = 1, \gamma(x_i,n) = 0\ \ ;\ \ \mu(x_i,n) = 0, \gamma(x_i,n) = 1\ \ ; \tag{6}$$

Consider, $|f_n * \mu_{\tilde{A}}(x_i,n) - g_n * \gamma_{\tilde{A}}(x_i,n)| = \sigma_{\tilde{A}}(x_i,n)$

$$E_3(\tilde{A}) = \frac{1}{2N}\sum_{n=1}^{N}\left[e^{-\sigma_{\tilde{A}}(x_i,n)} - e^{-1} + (1 - \sigma_{\tilde{A}}(x_i,n))\right]$$

From Eq. (6), it is obvious that $\sigma_{\tilde{A}}(x_i,n) = 1$ for either case when $N \to \infty$.

Thus, $E_3(\tilde{A}) = 0$ is proven.

A2: If $E_3(\tilde{A}) = 1$, then considering the condition $\mu_{\tilde{A}}(x_i, n) = \gamma_{\tilde{A}}(x_i, n)$ such that $\sigma_{\tilde{A}}(x_i, n) = 0$. Then, when $N \to \infty$

$$E_3(\tilde{A}) = \frac{1}{2N}\sum_{n=1}^{N}\left[e^{-0} - e^{-1} + (1-0)\right] = 1$$

A3: Straightforward.

A4: Similar to the previous theorem. □

3.3. Entropy measure based on logarithmic function

Definition 3.6. If $A = \{((x_i, n), \mu(x_i, n), \gamma(x_i, n))\}$ is an IDFS, then the entropy measure based on the logarithmic function (E_3) is defined by the following Eq. (7)

$$E_3(\tilde{A}) = 1 - \frac{2}{3mN \times \log\frac{3}{2\sqrt{2}}}\sum_{i=1}^{m}\sum_{n=1}^{N} \times \left[\frac{(f_n * \mu_{\tilde{A}}(x_i, n) + g_n * \gamma_{\tilde{A}}(x_i, n) + 2)(f_n * \mu_{\tilde{A}}(x_i, n) - g_n * \gamma_{\tilde{A}}(x_i, n))^2}{(f_n * \mu_{\tilde{A}}(x_i, n) + 1)(g_n * \gamma_{\tilde{A}}(x_i, n) + 1)} \times \log\left(\frac{(f_n * \mu_{\tilde{A}}(x_i, n) + g_n * \gamma_{\tilde{A}}(x_i, n) + 2) + 2}{2\sqrt{(f_n * \mu_{\tilde{A}}(x_i, n) - g_n + 1)(\gamma_{\tilde{A}}(x_i, n) + 1)}}\right)\right] \quad (7)$$

Theorem 3.7. *The $E_3(\tilde{A})$ defined in Eq. (7) is an IDF entropy if it satisfies the Definition 3.1 .*

Proof. If a mapping satisfies the axiomatic properties listed in Definition 3.1, it can be classified as an entropy measure. Therefore, the axioms for Eq. (7) are validated here.

A1: If $E_3(\tilde{A}) = 0$, then when $n \to \infty$

$$E_3(\tilde{A}) = 1 - \frac{2}{3\log\frac{3}{2\sqrt{2}}}\sum_{n=1}^{N}\left[\frac{3}{2}\log\left(\frac{3}{2\sqrt{2}}\right)\right] \approx 0$$

At $(0, 1)$ or $(1, 0)$ the N values are considered to be tending to infinity as the relation should hold the condition. Thus, the relation holds at $(0, 1)$ or $(1, 0)$. The converse is straightforward.

A2: If $f_n \times \mu_{\tilde{A}}(x_i, n) = g_n \times \gamma_{\tilde{A}}(x_i, n) = 0$, then

$$E_3(\tilde{A}) = 1 - \frac{2}{3\log\frac{3}{2\sqrt{2}}}\left[\frac{0}{1}\log\left(\frac{2}{2\sqrt{1}}\right)\right] = 1$$

Thus the relation works only for $f_n \times \mu_{\tilde{A}}(x_i, n) = g_n \times \gamma_{\tilde{A}}(x_i, n)$. The converse is straightforward.

A3: Straightforward

A4: If $\tilde{A}$ and B are two IDFS, $E(\tilde{A}) \le E(\tilde{B})$, the conditions, $\mu_{\tilde{A}}(x_i, n) + \gamma_{\tilde{A}}(x_i, n) \ge \mu_{\tilde{B}}(x_i, n) + \gamma_{\tilde{B}}(x_i, n)$

Then, $E(\tilde{A}) \le E(\tilde{B})$. Hence it is straightforward. □

Example 1. The extended entropy measures from Eqs. (4), (5) and (7) are applied to several examples for $N = 1, 5, 10, 100, 1000, 10000$ values. From the example, the drawbacks such as counter-intuitive and unreasonable results, the zero divisor problem, violation of entropy axioms and violation of crisp properties are validated and tabulated in Table 1.

From the Table 1, the axioms are substituted and explained as follows.

(i) According to the first axiom, the set $\tilde{A}$ is considered and thus, $E(\tilde{A}) = 0$ is only achieved in E_1 and E_2, when N values are found to be $1000, 10000, 100000$.

(ii) According to the second axiom, the set $\tilde{B}$ is considered and thus, $E(\tilde{B}) = 1$ is only achieved in E_1, when N values are found to be $1000, 10000, 100000$.

(iii) According to the third axiom, the sets $\tilde{E}$ and $\tilde{E}^c$ are considered and thus, $E(\tilde{E}) = E(\tilde{E}^c)$ is only achieved in E_1 and E_2, when N values are found to be $10000, 100000$.

(iv) According to the fourth axiom, the sets $\tilde{C}$ and $\tilde{D}$ are considered and thus, $E(\tilde{D}) \le E(\tilde{C})$ is achieved in all three E_1, E_2 and E_3 entropy measures, for all N values.

Thus, the set of examples defined in Table 1 explains the drawbacks faced by the entropy measures E_1, E_2 and E_3 and shows that E_3 undergoes counter-intuitive and unreasonable results, zero divisor problems, violation of entropy axioms and violation of crisp properties. In Table 2,

Table 1
Examples of the IDFE methods.

Sets		$\tilde{A}$	$\tilde{B}$	$\tilde{C}$	$\tilde{D}$	$\tilde{E}$	$\tilde{E}^c$	$\tilde{F}$
μ_i		$\mu_1 = 1$	$\mu_2 = 0.5$	$\mu_3 = 0.8$	$\mu_4 = 0.6548$	$\mu_5 = 0.7693$	$\mu_6 = 0.2307$	$\mu_7 = 0.5$
γ_i		$\gamma_1 = 0$	$\gamma_2 = 0.5$	$\gamma_3 = 0.2$	$\gamma_4 = 0.3452$	$\gamma_5 = 0.2307$	$\gamma_6 = 0.7693$	$\gamma_7 = 0.3$
N values	**Methods**	$\mu + \gamma = 1$	$\mu = \gamma$	$\mu > \gamma$	$\mu > \gamma$	$\mu > \gamma$	$\mu < \gamma$	$0 \le \mu + \gamma \le 1$
$N = 1$	E_1	0.1500	0.8000	0.5700	0.8749	0.6345	0.2345	0.9500
	E_2	0.2442	1.5132	1.0171	1.6889	1.1501	0.3883	1.8728
	E_3	−0.9747	−2.6745	0.5083	0.9804	1.5921	−19.425	0.3434
$N = 5$	E_1	0.0500	0.9333	0.4567	0.7519	0.5191	0.3858	0.8500
	E_2	0.0798	1.8313	0.7935	1.4043	0.9151	0.6597	1.6296
	E_3	−1.2573	−0.7262	1.4492	1.5636	1.6270	−12.726	0.8397
$N = 10$	E_1	0.0273	0.9636	0.4309	0.7239	0.4929	0.4201	0.8273
	E_2	0.0434	1.9071	0.7444	1.3423	0.8026	0.7240	1.5763
	E_3	−1.3264	−0.3765	1.4122	1.6471	1.6071	−11.431	0.9187
$N = 100$	E_1	0.0030	0.9960	0.4034	0.6941	0.4648	0.4569	0.8030
	E_2	0.0047	1.9898	0.6925	1.2771	0.8092	0.7940	1.5201
	E_3	−1.4023	−0.0388	1.3634	1.7172	1.5747	−10.1346	0.9898
$N = 1000$	E_1	2.997×10^{-04}	0.9996	0.4003	0.6908	0.4617	0.4609	0.8003
	E_2	4.7415×10^{-04}	1.9990	0.6862	1.2700	0.8033	0.8017	1.5139
	E_3	−1.4107	−0.0039	1.3574	1.7137	1.5704	−9.9976	0.9968
$N = 10000$	E_1	2.999×10^{-05}	1	0.4000	0.6904	0.4614	0.4614	0.8000
	E_2	4.7455×10^{-05}	1.9999	0.6863	1.2693	0.8027	0.8025	1.5133
	E_3	−1.4116	-3.8876×10^{-04}	1.3568	1.7243	1.5700	−9.9839	0.9975
$N = 100000$	E_1	3.00×10^{-06}	1	0.4000	0.6904	0.4614	0.4614	0.8000
	E_2	4.7459×10^{-06}	2	0.6862	1.2692	0.8026	0.8026	1.5132
	E_3	−1.4117	-3.8877×10^{-05}	1.3567	1.7244	1.5700	−9.9825	0.9976

Table 2
Comparative evaluation of entropy measures under varying N.

Case	N	Input Condition	E_1	E_2	E_3	Observation
1	1	General case	Stable (within [0, 1])	> 1 (violates bound)	Negative	E_2 and E_3 inconsistent
2	10	Increasing N	Converging	Still > 1	Negative persists	Instability continues
3	100	Large N	Near 0 (satisfies crisp condition)	Near 0	Negative / undefined	E_3 violates non-negativity
4	1000	Very large N	→ 0 (consistent)	→ 0 (slow convergence)	Irregular	E_1 shows best stability

a concise comparative analysis of entropy measures under varying values of N is given. Hence, the results claim that E_1 performs better when compared with the other defined entropy measures.

4. Methodology

This section outlines the methodology for determining the shortest path through the application of the modified BFS algorithm under the IDF environment and optimizing it through the multi-objective intuitionistic dense fuzzy optimization model (MOIDFOM). Additionally, prioritizing the robots is achieved using a novel MCDM technique within the framework of IDFS.

4.1. Multi-objective intuitionistic dense fuzzy optimization model

In this subsection, a MOIDFOM is defined and the solution approach proposed by Jimenez Lopez [36] is integrated to address the optimization model. A description of the optimization model is shown below.

Symbol	Description
N	Set of nodes in the operational network
A	Set of arcs connecting nodes
i, j	Node indices
s, t	Source and destination nodes
x_{ij}	Binary decision variable indicating whether arc $i \to j$ is selected
$\tilde{d}_{ij}$	Intuitionistic dense fuzzy distance value
$\tilde{r}_{ij}$	Intuitionistic dense fuzzy risk value
ρ	Importance weight of distance
β	Importance weight of risk
m	Number of criteria
N	Sequence parameter in IDFS
μ	Membership degree
γ	Non-membership degree

Decision variables:

$$x_{ij} : \begin{cases} 1 & , \quad arc(i,j) \\ 0 & , \quad \text{Otherwise} \end{cases} \tag{8}$$

In the MOIDFOM, the objective function Eq. (8) minimizes both the total risk and path distance values associated with the chosen path. This study aims to find a solution for identifying the safest shortest path. The objective function is optimized while considering the importance weights and relevant constraints associated with $(\widetilde{d_{ij}}, \widetilde{r_{ij}})$, as outlined in Eq. (9). The variable x_{ij} in this context signifies whether the arc (i, j) is utilized or not.

$$\min(Z) = \sum_i \sum_j \left((\rho) \frac{\tilde{d}_{ij}^{\mu} + \tilde{d}_{ij}^{\gamma}}{2} + (1-\beta) \frac{\tilde{r}_{ij}^{\mu} + \tilde{r}_{ij}^{\gamma}}{2} \right) (x_{ij}) \tag{9}$$

4.2. Multi-objective intuitionistic dense fuzzy breadth-first search algorithm

In this part, the MO-IDF-BFS algorithm is discussed and its pseudocode is provided in Algorithm 1.

Input: Let $G = (V, E)$ be a connected graph where V is the set of vertices and E is the set of edges.
STEP-1: Start by selecting a source node and a sink node to begin the search. This node can be any node in the graph.
STEP-2: Create an empty queue to keep track of visited nodes. Add the source node to the queue to indicate that it has been visited.
STEP-3: While the queue is not empty, do the following:

1. Dequeue a node from the front of the queue (current node).
2. If the current node is the goal node, the search is successful and you have found the shortest path.
3. Otherwise, explore the neighbors of the current node that are adjacent to it.
4. Enqueue the unvisited neighbors into the queue.
5. Continue the exploration process until the goal node is reached.

STEP-4: Once the queue is empty, it means that all the nodes reachable from the source node have been visited. If the sink node (the target node) has not been found, then there is no path from the source node to the sink node. In this case, the algorithm will return False or None, depending on the implementation.
STEP-5: Path Optimization

1. Once the goal is reached, consider the obtained path as the shortest path in the graph between the start and goal nodes.
2. Optimize the path using a MOIDFOM as given in Eq. (9). Thus, the optimized path is obtained.

4.3. AI based IDF decision making algorithm

Nowadays, AI is employed across various fields to achieve more optimized and accurate results. In the context of search algorithms, AI prioritizes outputs by analyzing network paths. This section comprises two stages. In stage 1, a modified IDF-BFS algorithm is developed using BFS and a multi-objective optimization model to identify the shortest path in a network G. Stage 2 involves MCDM models under IDFS to prioritize alternatives based on multiple criteria. The criteria for the MCDM problems are determined using the shortest path identified in stage 1.
STAGE-1: MO-IDF-BFS algorithm
The shortest path in the network $G = (V, E)$ where V and E are set of vertices and edges is computed using the novel BFS algorithm, as detailed in Section 4.2.
STAGE-2: IDF-MCDM algorithm
Step-1: Formulate the linguistic decision matrix (LD).
The linguistic variables from the experts are assigned to frame the linguistic decision matrix for m alternatives and n criteria.

$$LD = \left[x_i c_j\right]_{m\times n} = \begin{array}{c} \\ A_1 \\ A_2 \\ \vdots \\ A_n \end{array} \begin{array}{c} \begin{array}{cccc} C_1 & C_2 & \cdots & C_n \end{array} \\ \begin{pmatrix} x_1c_1 & x_1c_2 & \cdots & x_1c_n \\ x_2c_1 & x_2c_2 & \cdots & x_2c_n \\ \vdots & \vdots & \ddots & \vdots \\ x_mc_1 & x_mc_2 & \cdots & x_mc_n \end{pmatrix} \end{array}$$

where $x_i c_j$ represents the linguistic variable for the ith alternative with respect to the jth criterion.
Step-2: Convert LD into an intuitionistic dense fuzzy matrix (D).

Algorithm 1 Pseudocode for multi-objective intuitionistic dense fuzzy BFS algorithm.

function BFS MULTIPLE SHORTEST PATHS((V, E), SOURCES, SINKS)
  for each vertex in V **do** ▷ *Initialize all nodes as unvisited and set distance to infinity*
    mark vertex with 0
    predecessor[vertex] ← []
    init(queue) ▷ *create empty queue*
    for each source in sources **do**
      mark source with 1
      distance[source] ← 0
      inject(queue, source) ▷ *enqueue each source node*
      all paths ← [] ▷ *List to store all paths to sink nodes*
    end for
  end for
  while queue is non-empty **do** ▷ *Perform BFS*
    u ← eject(queue) ▷ *dequeue u*
    for each edge (u, w) adjacent to u **do**
      if w is marked with 0 **then** ▷ *if w is unvisited*
        mark w with 1 ▷ *mark w as visited*
        distance[w] ← distance[u] + 1 ▷ *update distance*
        predecessor[w] ← [u] ▷ *set predecessor of w to u*
        inject(queue, w) ▷ *enqueue w*
      else if distance[w] == distance[u] + 1 **then**
        predecessor[w].append(u) ▷ *add u as another predecessor of w*
        if w in sinks and w not in all paths **then**
          all paths.append(reconstruct path(predecessor, sources, w))
        end if
      end if
    end for
  end while
end function
function RECONSTRUCT PATH (PREDECESSOR, SOURCES, SINK)
  paths ← [[]] ▷ *list of paths to be built*
  current paths ← [[sink]] ▷ *starting with the sink node*
  while any(path[0] not in sources for path in current paths) **do**
    new paths ← []
    for path in current paths **do**
      if path[0] in sources **then**
        paths.append(path)
      else
        for pred in predecessor[path[0]] **do**
          new path ← [pred] + path
          new paths.append(new path)
          current paths ← new paths
        end for
      end if
    end for
  end while
end function
valid paths ← [path in paths if path[0] in sources] ▷ Only keep valid paths that start with a source node
return valid paths
Optimize the valid paths by MOIDFOM ▷ Path with minimum value is considered as shortest path

$$\min(Z) = \sum_i \sum_j \left((\rho) \frac{\tilde{d}_{ij}^{\mu} + \tilde{d}_{ij}^{\gamma}}{2} + (1-\beta) \frac{\tilde{r}_{ij}^{\mu} + \tilde{r}_{ij}^{\gamma}}{2} \right) (x_{ij})$$

Table 3
ITpDF linguistic scale.

Importance level	IDFS
Very Low (VL)	$(0.05 * f_n,\ 0.75 * g_n)$
Low (L)	$(0.25 * f_n,\ 0.55 * g_n)$
Medium (M)	$(0.45 * f_n,\ 0.35 * g_n)$
High (H)	$(0.65 * f_n,\ 0.15 * g_n)$
Very High(VH)	$(0.85 * f_n,\ 0.05 * g_n)$

The decision makers (q) may decide on the linguistic terminology for the LD. In this case, the weighting of the decision makers based on their background, place in the industry, and other factors is determined by the n values. Thus, using the IDFS linguistic values from Table 3, LD is converted into (D) where the linguistic values can be converted into IDFS.

$$D = \left[l_{ij}\right]_{m\times n} = \begin{pmatrix} l_{11} & l_{12} & \cdots & l_{1n} \\ l_{21} & l_{22} & \cdots & l_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ l_{m1} & l_{m2} & \cdots & l_{mn} \end{pmatrix}$$

where $l_{ij} = (b, c, a = b - f_n, d = c + g_n; b, c, a' = b - f'_n, d' = c + g'_n)_{ij}$ represents the IDFN of the ith alternative with respect to the jth criterion.

Step-3:: Obtain the weight vector for the criteria W using the entropy measure $E(\tilde{A})$.

The entropy method $E(\tilde{A})$ can be used to determine the entropy values e_j of the jth factor given in Eq. (4). The weight vector of the criteria W of the jth factor can be obtained using Eq. (10)

$$W = [w_j]_{1\times n} = \frac{1 - e_j}{\sum_{i=1}^{n} 1 - e_j} \tag{10}$$

Step-4: Attain the normalized decision matrix (N).

In order to ensure comparability among the values, it is necessary to normalize the decision matrix (D) using Eq. (11).

$$\hat{x}_{ij} = \begin{cases} \dfrac{x_{ij}}{\max_i(x_{ij})} & \text{for beneficial criteria} \\ \dfrac{\min_i(x_{ij})}{x_{ij}} & \text{for cost criteria} \end{cases} \tag{11}$$

Consequently, the decision matrix D can be converted into a normalized decision matrix (N).

Step-5: Derive the four utility values from the Eqs. (12)–(15)

$$U_i^{wsd} = \sum_{j\in\Omega_{\max}} \hat{x}_{ij} w_j - \sum_{j\in\Omega_{\min}} \hat{x}_{ij} w_j \tag{12}$$

$$U_i^{wpd} = \prod_{j\in\Omega_{\max}} \hat{x}_{ij} w_j - \prod_{j\in\Omega_{\min}} \hat{x}_{ij} w_j \tag{13}$$

$$U_i^{wsr} = \frac{\sum_{j\in\Omega_{\max}} \hat{x}_{ij} w_j}{\sum_{j\in\Omega_{\min}} \hat{x}_{ij} w_j} \tag{14}$$

$$U_i^{wpr} = \frac{\prod_{j\in\Omega_{\max}} \hat{x}_{ij} w_j}{\prod_{j\in\Omega_{\min}} \hat{x}_{ij} w_j} \tag{15}$$

where U_i^{wsd} and U_i^{wpd} denote the difference between the influence of beneficial and non-beneficial criteria and U_i^{wsr} and U_i^{wpr} denote the ratio

between the influence of beneficial and non-beneficial criteria based on weighted sum (WS) and weighted product (WP) models, respectively.

Step-6: Recalculate the four utility values from the Eqs. (16)–(19)

$$\bar{U}_i^{wsd} = \frac{U_i^{wsd}}{1 + U_{\max_i}^{wsd}} \tag{16}$$

$$\bar{U}_i^{wpd} = \frac{U_i^{wpd}}{1 + U_{\max_i}^{wpd}} \tag{17}$$

$$\bar{U}_i^{wsr} = \frac{U_i^{wsr}}{1 + U_{\max_i}^{wsr}} \tag{18}$$

$$\bar{U}_i^{wpr} = \frac{U_i^{wpr}}{1 + U_{\max_i}^{wpr}} \tag{19}$$

where $\bar{U}_i^{wsd}$, $\bar{U}_i^{wpd}$, $\bar{U}_i^{wsr}$ and $\bar{U}_i^{wpr}$ are normalized values of U_i^{wsd}, U_i^{wpd}, U_i^{wsr} and U_i^{wpr}.

Step-7: Defuzzify the values of the alternatives.

Defuzzify the values of $\bar{U}_i^{wsd}$, $\bar{U}_i^{wpd}$, $\bar{U}_i^{wsr}$ and $\bar{U}_i^{wpr}$ to $\bar{D}_i^{wsd}$, $\bar{D}_i^{wpd}$, $\bar{D}_i^{wsr}$ and $\bar{D}_i^{wpr}$ using Eq. (3).

Step-8: Calculate the overall utility D of each alternative using Eq. (20)

$$D = \frac{1}{4}\left(\bar{D}_i^{wsd} + \bar{D}_i^{wpd} + \bar{D}_i^{wsr} + \bar{D}_i^{wpr}\right) \tag{20}$$

Step-9: Order the alternatives and choose the most appropriate one.

The alternatives are ranked in descending order, with the highest value indicating the most preferred option.

The MO-IDF-BFS method follows the same exploration pattern as the traditional breadth-first search algorithm. For a graph with $|V|$ vertices and $|E|$ edges, the traversal requires $O(V+E)$ time. During the traversal, additional computations are carried out to determine the fuzzy distance and risk for each edge. If k feasible paths are determined, the multi-objective evaluation step takes $O(k)$ time. The proposed IDFE–WISP model depends on m alternatives and n criteria, and its time complexity is $O(mn)$. Therefore, the overall time complexity of the proposed system is $O(V + E + mn)$. The overall flowchart of the method is given in Fig. 1

5. Application in fire rescue operations

Firefighting refers to the practice of extinguishing fires, rescuing people and animals and preventing the spread of fire and smoke to minimize damage to property and loss of life. Firefighters are trained and equipped to respond to various types of fires, accidents and emergencies. It is a challenging and demanding profession that requires courage, physical fitness, teamwork and a commitment to public service. They work under hazardous conditions and often put themselves at risk to protect others and save lives.

Integrating robots into firefighting operations offers several advantages, enhancing the overall effectiveness and safety of emergency response efforts. They may not replace human firefighters entirely, but they can complement human capabilities in various ways. The selection of robots to help humans in firefighting is going to be a difficult task as humans bring a combination of physical abilities, cognitive skills and emotional intelligence. Therefore, employing robots for human protection necessitates meeting specific criteria, such as selecting the most efficient path and optimizing risk and distance factors using advanced optimization models and MCDM to prioritize the robots and identify the most suitable one among them.

5.1. Illustration on fire rescue operation in mall

Consider a mall spanning a total area of 205,023 sq.ft., constructed with a main area covering 150,000 sq.ft. (excluding parking). The fire

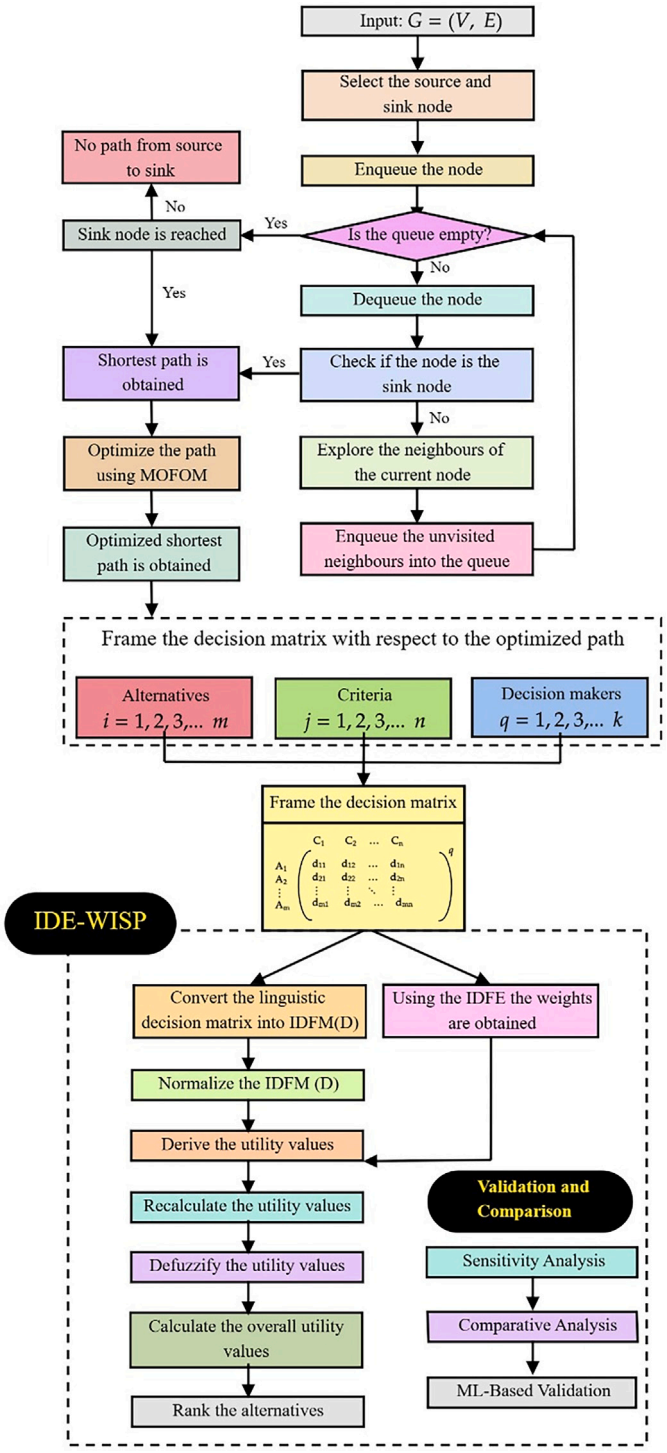


Fig. 1. Flowchart of the proposed method.

originated in the gaming zone, which occupies an area of 1150 sq.ft. as considered. A visual representation of this situation is illustrated in Fig. 2 where the path is defined in Fig. 3. Initially, the calculation of the fireload load is conducted to assess the quantity of combustible materials within the specified area. The combustible substance calculation is given in Table 4, where the fireload load is found to be 79,55,263 kcal/kg. This situation poses a significant danger to human firefighters. Consequently, the decision has been made to deploy robots to extinguish the fire.

Deploying robots in a challenging firefighting situation is a complex task. Therefore, the initial step involves determining the shortest path

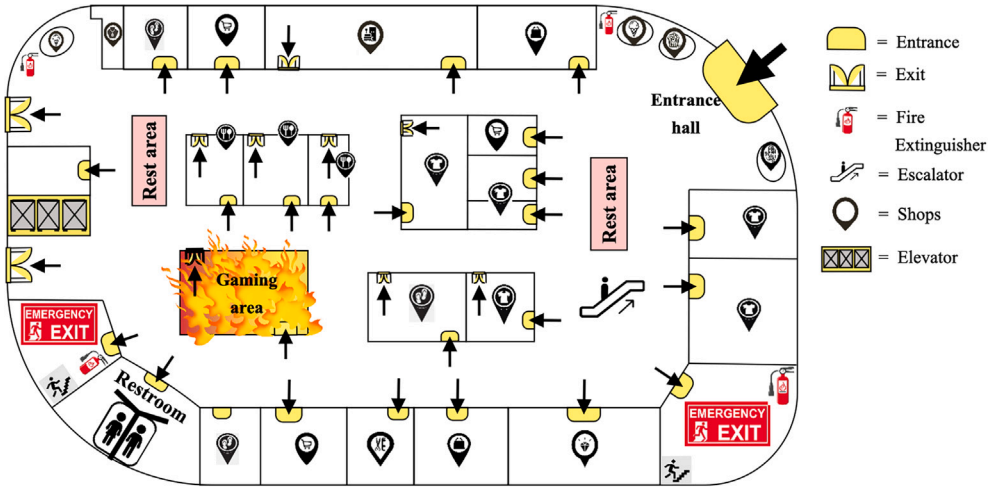


Fig. 2. Mall blue print.

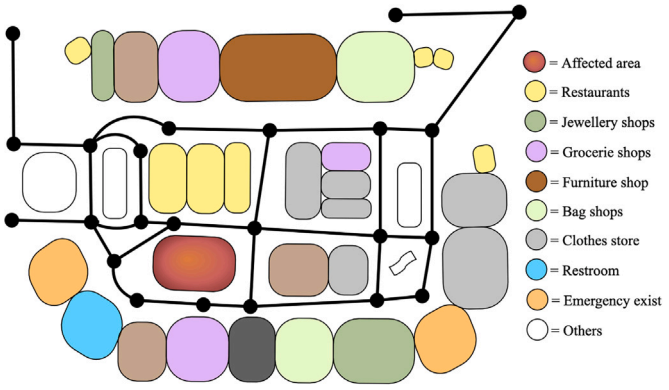


Fig. 3. Bubble diagram of the mall.

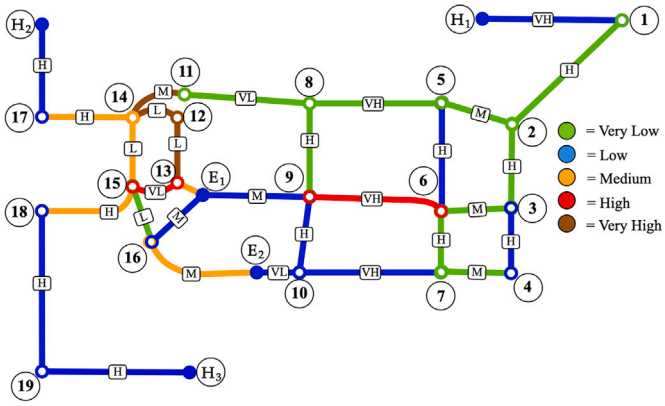


Fig. 4. Path graph.

by employing the MO-IDF-BFS technique to optimize risk and distance factors along the graph edges through advanced optimization models. Additionally, MCDM is applied to prioritize the robots, aiding in the identification of the most suitable one among them. Thus, the algorithm given in Section 4.3 is utilized.

Graph modeling and arc attributes: The scenario is based on reasonable assumptions. In which, the distances between nodes (arcs) and the associated risk levels (represented by colours or values) were derived using expert judgment. The distances reflect the relative positions of different zones within the mall, while risk levels were assigned considering fire safety guidelines and the calculated fireload for each area. This provides a hypothetical yet practical framework for evaluating robot deployment strategies.

Table 4
Total fireload.

Materials	Calorific value		Quantity in kg	Area in sq. ft	Total fireload
	MJ/kg	KJ/kg			
Plastic	20	25,000	18,000	1150	93,524
Wood	17.5	17,5000	25,000	1150	9,09,260
Papers	20	20,000	1000	1150	4157
Electrical items and circuits	3.598	3598	2580	1150	69,40,670
Miscellaneous	16.736	16,736	2200	1150	7652
					Total = 79,55,263

Table 5
Preference table for distance and risk levels.

Linguistic terms	Distance and risk values
VL	(0.1459, 0.8886)
L	(0.3405, 0.6795)
M	(0.5350, 0.4705)
H	(0.7295, 0.2614)
VH	(0.9241, 0.0523)

Relation of fireload to edge risks: The calculated fireload values (in kcal/kg) were used to measure the relative hazard of each path. Zones with higher fireload were noted for higher risk levels, which ensures that the path-planning algorithm takes into account both the physical distance and the potential danger. This helps guide robots along safer routes and minimizes their exposure to high-risk areas.

STAGE-1: BFS Algorithm

In the network $G(24, 31)$ given in Fig. 4, the information on fuzzy distance is provided on the corresponding arcs in terms of linguistic expressions. Additionally, different colours represent the risk levels, with their values tabulated in Table 5.

The shortest path of the network $G(24, 31)$ from the Fig. 4 is calculated using the BFS algorithm, which is detailed in Section 4.2. From the network $G(24, 31)$, the source nodes H_1, H_2 and H_3 represent three

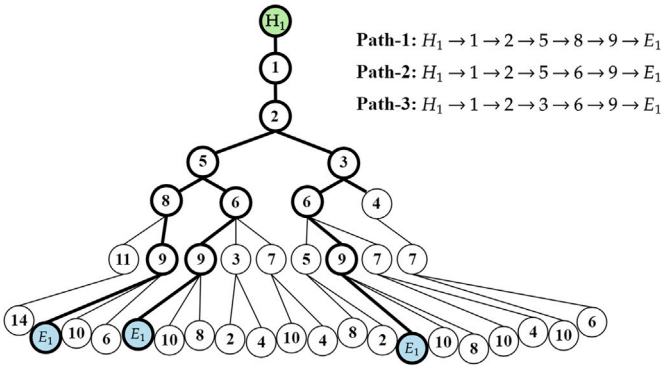


Fig. 5. Search tree diagram from H_1.

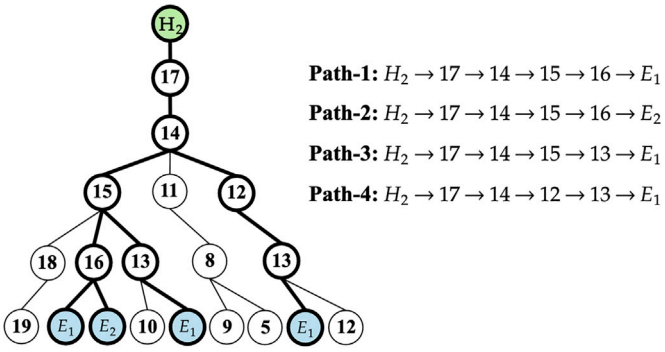


Fig. 6. Search tree diagram from H_2.

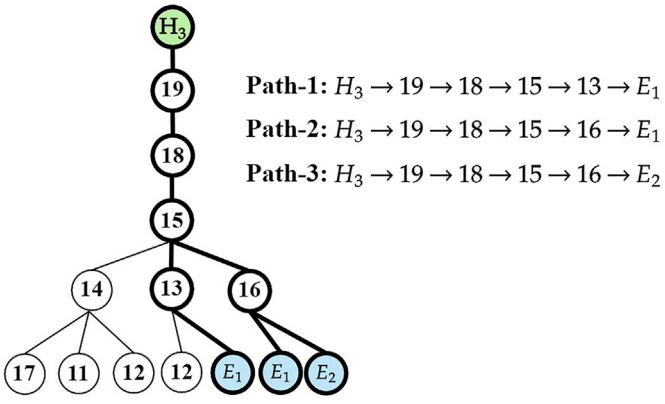


Fig. 7. Search tree diagram from H_3.

hydrants where the robots are connected to the hydrants and the sink nodes E_1 and E_2 represent the two entrances of the firing room. Thus, the tree structure obtained from the graph by performing the BFS algorithm is shown in Figs. 5–7.

The paths obtained from H_1, H_2 and H_3 to E_1 and E_2 are optimized using MOIDFOM given in Eq. (9) and are tabulated in Table 6. Hence, the shortest paths are prioritized and the best one is found to be $H_2 \rightarrow 17 \rightarrow 14 \rightarrow 12 \rightarrow 13 \rightarrow E_1$ and its value is 5.0128 which is considered to be the shortest path with less risk comparatively.

STAGE-2: IDFE-WISP method

In this stage, the most appropriate robot will be identified to assist firefighters in safely extinguishing the fire from a secure distance. Since the available robots differ in their technical capabilities, a structured evaluation is required to determine the most effective option.

Table 6
Optimized values.

Source node	Path no.	Path values	Optimal path values
H1	1	6.0618	6.0328
	2	6.0328	
	3	6.0400	
H2	1	5.0564	5.0128
	2	5.0492	
	3	5.0419	
	4	5.0128	
H3	1	5.0346	5.0346
	2	5.0491	
	3	5.0412	

Table 7
Evaluation criteria for firefighting robot selection.

Criterion	Definition	Unit	Direction
Nozzle Flow	Water discharge per minute	gpm	Beneficial
Robot Size	Dimensions	cm	Cost
Weight	Total mass	kg	Cost
Communication Range	Maximum operational distance	m	Beneficial
Speed	Maximum speed	km/h	Beneficial
Camera System	Number/quality of cameras	count/resolution	Beneficial
Maximum Water Stream	Maximum reach of water	m	Beneficial
Slope Capability	Maximum incline the robot can navigate	%	Beneficial
Operational Time	Maximum continuous operation	hrs	Beneficial
Degree of Freedom	Number of independent movements	°	Beneficial
Cost	Purchase price	$	Cost

Robot selection criteria: To ensure consistency in evaluation, the criteria used for comparing the robots are defined in Table 7, along with their measurement units and their nature (benefit or cost). This classification enables proper normalization and allows meaningful comparison among alternatives. The performance data of the robots with respect to these criteria are given in Table 8. These values form the basis for ranking the alternatives using the proposed decision-making approach. The presumptions provided in Table 9 represent the minimum operational requirements specified by the decision maker. These include constraints related to speed, battery life, slope handling capability and other functional limits. Such conditions act as feasibility filters, ensuring that only practically deployable robots are considered for evaluation. By separating these baseline requirements from the evaluation criteria, redundancy is avoided and clarity is maintained in the decision process. The ranking of robots is therefore based solely on measurable performance attributes, while the presumptions ensure operational suitability in real firefighting conditions.

The challenge here is to prioritize the robot to support firefighters, enabling them to safely extinguish the fire from a secure distance. Thus, the defined algorithm is used to identify the better-performing robots by analyzing their criteria values, displayed in Table 8 and the presumption of the decision maker is provided in Table 9.

Step-1: The consolidated opinions of decision-makers are represented as LD and are given in Table 10

Step-2: The collective viewpoints of decision-makers, where the n values of the IDFS linguistic scale fluctuate based on the experience level of the decision-makers.

The n values increase according to the experience of the decision maker (DM) in the field. Three DMs are taken into consideration: the first, D1, has five years of experience; the second, D2, has thirteen years and the third, D3, has thirty-five years. Hence, the n values are taken to

Table 8
Robots and their criteria.

	Nozzle Flow (gpm)	Size (cm)	Wt (kg)	Range (m)	Speed (km/hr)	Cam
Thermite RS1	1250	196.215× 111.76 × 162.56	725 kg	300–500	9.7	2
Thermite RS3	2500	304×166.624×164.465	1587.6	300–500	9.7	2
Colossus	H	160×78×76	485	200–500	3.5	2
TAF60	1585	311×165×1999	4000	300	9	2
TAF60X	1585	314×165×216	4030	300	9 km /hr	2
TAF35	1241	300×165×220	3900	3900	9	2
Hercules	3700	H	3900	3900	H	2
Rhino	1000	H	3900	3900	H	2

	Max.stream (m)		Slope (%)		Time (hrs)	DOF (∘)	Cost$
	Horizontal	Vertical	Max. grade slope	Max. side slope			
Thermite RS1	100	50	50	50	20	M	$98,500
Thermite RS3	100	50	35	35	20	M	$278,000
Colossus	H	M	50	45	6–8	7	$180,600
TAF60	80	40	30	27	M	M	H
TAF60X	80	40	30	27	M	M	H
TAF35	80	40	58	27	5–7	M	H
Hercules	70	40	30	35	3–4	360	M
Rhino	80	40	30	35	5–8	360	M

Table 9
Presumptions of the decision maker.

NF (gpm)	SIZE (cm)	Wt(kg)	Range (m)	SP (km/hr)	Cam	M.strm HR(m)
1000 gpm	200 × 100 × 80	2000 kg	300 m	9 km	2	80
M.strm VT(m)	M.slope GRD(%)	M.slope SIDE(%)	Time(hrs)	DOF(∘°)	Cost($)	
40	30 %	27 %	7 hrs	5	$150,000	

Table 10
Opinions of the decision makers.

	Nozzle-Flow	Size (cm)	Wt	Range	Speed	Cam	Max.stream		Slope		Time	DOF	Cost
							Horizoltal	Vertical	Max. grade slope	Max. side slope			
Thermite RS1	M	H	VL	H	M	M	H	H	VH	VH	VH	M	VL
Thermite RS3	H	H	L	H	VH	M	H	H	H	H	VH	M	VH
Colossus	H	VL	VL	VH	VL	M	H	M	VH	VH	H	H	M
TAF60	M	H	H	M	M	M	M	M	M	M	M	M	H
TAF60X	M	H	VH	M	M	M	M	M	M	M	M	M	H
TAF35	L	H	VH	M	M	M	M	M	VH	M	M	M	H
Hercules	VH	H	VH	M	H	M	L	M	M	H	VL	VH	M
Rhino	M	H	VH	M	H	M	L	M	M	H	H	VH	M

Table 11
Linguistic table.

	$n = 10$	$n = 100$	$n = 1000$
VL	(0.1459, 0.8886)	(0.1496, 0.8542)	(0.1500, 0.8504)
L	(0.3405, 0.6795)	(0.3490, 0.6532)	(0.3499, 0.6503)
M	(0.5350, 0.4705)	(0.5484, 0.4522)	(0.5498, 0.4502)
H	(0.7295, 0.2614)	(0.7478, 0.2512)	(0.7498, 0.2501)
VH	(0.9241, 0.0523)	(0.9472, 0.0502)	(0.9497, 0.0500)

be $n = 10, 100, 1000$. As a result, linguistic terms are converted into an IDFS to build (D) utilizing the values from Table 11.

Step-3: By calculating the total preference value using Eq. (10), the objective weights for each criterion are determined. Hence, the weights are tabulated in Table 12.

Step-:4 The normalized decision matrix (N) is attained using Eq. (11) and the matrix N is given in Table 13.

Step-5: The four utility values from U_i^{wsd}, U_i^{wsr}, U_i^{wpd} and U_i^{wpr} are calculated and displayed in Table 14.

Step-6: The obtained four utility values from Table 14 are recalculated using Eq. (16), (17), (18), and (19), where $\bar{U}_i^{wsd}$, $\bar{U}_i^{wpd}$, $\bar{U}_i^{wsr}$ and $\bar{U}_i^{wpr}$ are the normalized values of U_i^{wsd}, U_i^{wpd}, U_i^{wsr} and U_i^{wpr} and the values are tabulated in Table 15.

Step-7: The values of $\bar{U}_i^{wsd}$, $\bar{U}_i^{wpd}$, $\bar{U}_i^{wsr}$ and $\bar{U}_i^{wpr}$ are defuzzified using Eq. (3). The defuzzified values are tabulated in Table 15.

Step-8: The overall utility D of each alternative is calculated and tabulated in Table 16

Step-9: The score values of the alternatives are ordered in descending order and the alternative obtaining the first rank is considered the most favorable. Hence, the prioritization of the alternatives is displayed in Table 16.

5.2. Uncertainty handling and robustness analysis

Decision-making problems involving expert assessments inherently contain uncertainty arising from subjective judgments, modeling assumptions, and environmental variability. Therefore, it is necessary to examine whether the proposed decision framework maintains stable results when small variations occur in the input information. To address this issue, a robustness analysis was carried out by introducing controlled perturbations into the linguistic decision matrix and observing the resulting changes in the ranking of alternatives.

Table 12
Weights of the criteria.

NF (gpm)	SIZE (cm)	Wt(kg)	Range (m)	SP (km/hr)	Cam	M.strm HR(m)
0.0538	0.1225	0.1395	0.0732	0.0779	0.0193	0.0520
M.strm VT(m)	M.slope GRD(%)	M.slope SIDE(%)	Time(hrs)	DOF(∘°)	Cost($)	
0.0377	0.1109	0.0756	0.0883	0.0670	0.0822	

Table 13
Normalized decision matrix.

	Nozzle Flow	Size (cm)	Wt	Range	Speed	Cam	
Thermite RS1	(0.5789, 9.0013)	(4.9991, 0.2941)	(1.0000, 1.0000)	1.0000, 1.0000	0.5789, 9.0026	1.0000, 1.0000	
Thermite RS3	(0.7895, 5.0013)	(4.9991, 0.2941)	(2.3331, 0.7647)	1.0000, 1.0000	1.0000, 1.0000	1.0000, 1.0000	
Colossus	(0.7895, 5.0013)	(1.0000, 1.0000)	(1.0000, 1.0000)	1.0000, 1.0000	0.1579, 17.0046	1.0000, 1.0000	
TAF60	(0.5789, 9.0026)	(4.9991, 0.2941)	(4.9991, 0.2941)	0.5789, 9.0026	0.5789, 9.0026	1.0000, 1.0000	
TAF60X	(0.5789, 9.0026)	(4.9991, 0.2941)	(6.3322, 0.0588)	0.5789, 9.0026	0.5789, 9.0026	1.0000, 1.0000	
TAF35	(0.3685, 13.0033)	(4.9991, 0.2941)	(6.3322, 0.0588)	0.5789, 9.0026	0.5789, 9.0026	1.0000, 1.0000	
Hercules	(1.0000, 1.0000)	(4.9991, 0.2941)	(6.3322, 0.0588)	0.5789, 9.0026	0.7895, 4.8046	1.0000, 1.0000	
Rhino	(0.5789, 9.0026)	(4.9991, 0.2941)	(6.3322, 0.0588)	0.5789, 9.0026	0.7895, 4.8046	1.0000, 1.0000	
	Max.stream		Slope		Time	DOF	Cost
	Horizoltal	Vertical	Max. grade slope	Max. side slope			
Thermite RS1	(1.0000, 1.0000)	(1.0000, 1.0000)	(1.0000, 1.0000)	1.0000, 1.0000	1.0000, 1.0000	0.5789, 9.0026	(1.0000, 1.0000)
Thermite RS3	(1.0000, 1.0000)	(1.0000, 1.0000)	(0.7895, 4.8046)	0.7895, 4.8046	1.0000, 1.0000	0.5789, 9.0026	6.2801, 0.0588
Colossus	(1.0000, 1.0000))	(0.7333, 1.8738)	(1.0000, 1.0000)	1.0000, 1.0000	0.7895, 4.8046	0.7895, 4.8046	3.6358, 0.5294
TAF60	(0.7333, 1.8738)	(0.7333, 1.8738)	(0.6498, 7.6905)	0.5789, 9.0026	0.5789, 9.0026	0.5789, 9.0026	4.9579, 0.2825
TAF60X	(0.7333, 1.8738)	(0.7333, 1.8738)	(0.6498, 7.6905)	0.5789, 9.0026	0.5789, 9.0026	0.5789, 9.0026	4.9579, 0.2825
TAF35	(0.7333, 1.8738)	(0.7333, 1.8738)	(1.0000, 1.0000)	0.5789, 9.0026	0.5789, 9.0026	0.5789, 9.0026	4.9579, 0.2825
Hercules	(0.4667, 2.7064)	(0.7333, 1.8738)	(0.6498, 7.6905)	0.7895, 4.8046	0.1592, 17.0046	1.0000, 1.0000	3.6358, 0.5294
Rhino	(0.4667, 2.7064)	(0.7333, 1.8738)	(0.6498, 7.6905)	0.7895, 4.8046	0.7895, 4.8046	1.0000, 1.0000	3.6358, 0.5294

Table 14
Recalculated utility values.

	WSD	WSR	WPD	WPR
Thermite RS1	(0.3198, 0.1150)	(2.3176, 1.0642)	(7.0490, 7.0692)	(3.4981, 3.1015)
Thermite RS3	(0.4400, −2.0939)	(4.3777, 0.4250)	(7.3097, 6.5516)	(3.8302, 2.6927)
Colossus	(0.2285, 1.2272)	(1.7707, 2.2409)	(6.7925, 7.3156)	(3.3103, 3.2896)
TAF60	(0.3140, −0.4563)	(5.6823, 0.8967)	(7.1405, 7.1979)	(3.8582, 2.7910)
TAF60X	(0.3320, −1.0073)	(7.7671, 0.7973)	(7.2515, 7.1444)	(4.0377, 2.7543)
TAF35	(0.3570, −1.2303)	(8.2763, 0.7525)	(7.2654, 6.9825)	(4.0436, 2.7146)
Hercules	(0.3585, −1.3966)	(6.0568, 0.6994)	(7.1395, 6.8390)	(3.9135, 2.6944)
Rhino	(0.3897, −2.0586)	(6.4958, 0.5569)	(7.2951, 6.8060)	(3.9769, 2.6862)
MAX,MIN	(0.4400, −2.0939)	(8.2763, 0.4250)	(7.3097, 6.5516)	(4.0436, 2.6862)

Table 15
Scoring values.

	NWSD	NWSR	NWPD	NWPR	Defuzzify/ Score			
					NWSD	NWSR	NWPD	NWPR
Thermite RS1	(0.9197, 1.4121)	(0.4166, 1.5542)	(0.9676, 1.0228)	(0.8938, 1.0604)	0.2538	0.9977	−0.0688	0.7933
Thermite RS3	(1.0000, 1.0000)	(0.6400, 1.0000)	(0.9990, 1.0000)	(0.9595, 1.0000)	0.5000	0.6800	0.3200	0.5005
Colossus	(0.8710, 1.7635)	(0.3651, 2.3186)	(0.9490, 1.0499)	(0.8707, 1.0927)	0.0538	1.1992	−0.4768	1.1848
TAF60	(0.9154, 2.4684)	(0.6706, 1.5819)	(0.9740, 1.0854)	(0.9463, 1.0328)	−0.2765	1.3989	0.0443	0.8040
TAF60X	(0.9386, 2.3641)	(0.9532, 1.4604)	(0.9947, 1.0801)	(0.9986, 1.0236)	−0.2128	1.2055	0.2464	0.7328
TAF35	(0.9549, 2.1168)	(1.0000, 1.3668)	(0.9967, 1.0538)	(1.0000, 1.0090)	−0.0809	1.0584	0.3166	0.6850
Hercules	(0.9522, 1.8799)	(0.8150, 1.3085)	(0.9883, 1.0411)	(0.9829, 1.0060)	0.0362	1.0325	0.2532	0.6601
Rhino	(0.9726, 1.5612)	(0.8595, 1.1805)	(1.0000, 1.0430)	(0.9907, 1.0071)	0.2057	0.8509	0.3395	0.5902

The perturbed linguistic terms were converted into IDFNs using the same scale employed in the proposed method. A Monte Carlo experiment was conducted to investigate the stability of the obtained rankings. In each simulation, the linguistic decision matrix was randomly modified within ±1 linguistic scale while keeping the criteria structure unaltered. The complete evaluation procedure was then repeated for a large number of iterations ($i = 1000$) and the rankings were recorded for each simulation. The original ranking given in Table 16 was used as the reference ranking for evaluating the stability of the decision model. The frequency with which each robot appears in the first position during the simulations was calculated. A more comprehensive picture of ranking behaviour is shown in Table 17.

The results indicate that Thermite RS3 occupies the first position in approximately 50.1% of the simulations, followed by Rhino with a probability of 24.8%. The remaining alternatives achieve the top rank only occasionally. This observation suggests that Thermite RS3 remains the most competitive option even when moderate variations are introduced into the input evaluations. The rank-reversal rate (RRR) is 0.993,

Table 16
Utility values and their ranking.

	Overall utility value	Rank
Thermite RS1	0.4940	5
Thermite RS3	**0.5001**	1
Colossus	0.4903	8
TAF60	0.4927	7
TAF60X	0.4930	6
TAF35	0.4948	4
Hercules	0.4955	3
Rhino	0.4966	2

Table 17
Rank frequency distribution.

Alternative	R1	R2	R3	R4	R5	R6	R7	R8
Thermite RS1	0.001	0.008	0.021	0.034	0.067	0.192	0.362	0.315
Thermite RS3	0.501	0.181	0.101	0.092	0.094	0.022	0.008	0.001
Colossus	0.000	0.004	0.003	0.012	0.049	0.139	0.265	0.528
TAF60	0.074	0.145	0.169	0.202	0.206	0.111	0.063	0.030
TAF60X	0.086	0.154	0.216	0.222	0.182	0.079	0.043	0.018
TAF35	0.087	0.209	0.225	0.204	0.157	0.063	0.036	0.019
Hercules	0.003	0.018	0.059	0.102	0.162	0.364	0.209	0.083
Rhino	0.248	0.281	0.206	0.132	0.083	0.030	0.014	0.006

Table 18
Sensitivity analysis.

Robots	Max-min	Exp	log
Thermite RS1	0.4935	0.4951	0.4950
Thermite RS3	0.5000	0.5000	0.5000
Colossus	0.4894	0.4933	0.4932
TAF60	0.4916	0.4903	0.4902
TAF60X	0.4915	0.4900	0.4899
TAF35	0.4935	0.4910	0.4909
Hercules	0.4955	0.4941	0.4940
Rhino	0.4959	0.4935	0.4935

computed as $RRR = \frac{m}{i}$, where m and i represent the number of rank reversals and the total runs, respectively. This result reveals that changes in the input data can lead to different rankings of alternatives. However, these changes mainly occur among robots with very close utility values.

5.3. Sensitivity study

In this section, sensitivity analysis is employed to investigate the impact of the remaining two IDFE measures defined in this work given in Eqs. (5) and (7) in order to uncover variations in the entropy weight measure. The ranking of the robots is evaluated and summarized in Table 18 and visualized in Fig. 8.

5.4. Comparative study

In this section, four widely-used MCDM models namely WASPAS, COPRAS, WAS, WAP, and ARAS are compared with the proposed model. Thus, the score values obtained from all the approaches are compared with the values obtained from the proposed model and displayed in Table 19.

Since the rank of the alternatives concerning the four methods varies greatly, the Borda count method is utilized to determine the collective preferences among the group of ranks and the consolidated ranking is given in Table 20. Thus, by comparing the ranking of the Borda count method with the proposed method, it is found that the proposed ranking ranks first.

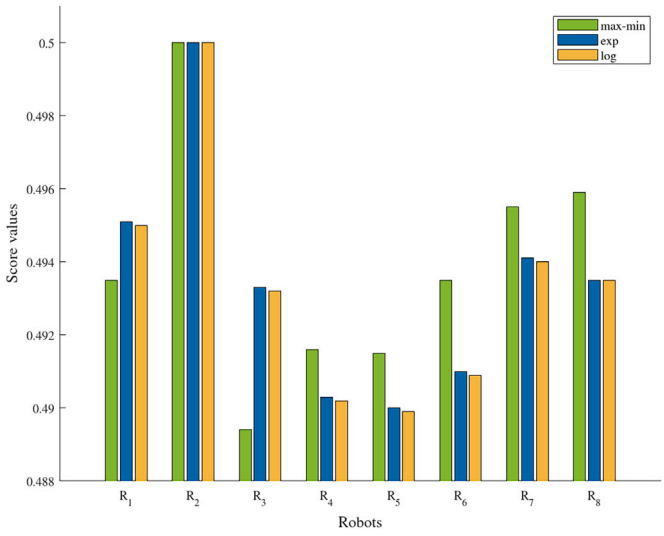


Fig. 8. Sensitivity analysis.

Table 19
Comparative analysis with other MCDM methods.

Robot	WASPAS		COPRAS		WAS	
	Score value	Rank	Score value	Rank	Score value	Rank
Thermite RS1	0.1115	2	0.0652	7	−0.1189	2
Thermite RS3	0.4697	1	0.1482	3	0.3122	1
Colossus	−0.1850	6	0.0446	8	−0.4723	4
TAF60	−0.5764	8	0.1400	6	−1.0792	8
TAF60X	−0.3885	7	0.1553	2	−0.9681	7
TAF35	−0.1823	5	0.1559	1	−0.7322	6
Hercules	−0.0646	4	0.1445	5	−0.5195	5
Rhino	0.1115	3	0.1464	4	−0.2160	3
Robot	WAP		ARAS		Proposed	
	Score value	Rank	Score value	Rank	Score value	Rank
Thermite RS1	0.3419	5	0.4940	5	0.1008	7
Thermite RS3	0.6273	1	0.5001	1	0.1564	2
Colossus	0.1023	7	0.4903	8	0.0966	8
TAF60	−0.0735	8	0.4927	7	0.1904	1
TAF60X	0.1910	6	0.4930	6	0.1248	5
TAF35	0.3676	4	0.4948	4	0.1245	6
Hercules	0.3902	3	0.4955	3	0.1310	4
Rhino	0.4390	2	0.4966	2	0.1312	3

Table 20
Comparative analysis ranking using Borda method.

ROBOT	Borda	Rank	Proposed
Thermite RS1	25	4	4
Thermite RS3	38	1	1
Colossus	12	7	8
TAF60	08	8	6
TAF60X	17	6	7
TAF35	25	4	5
Hercules	26	3	3
Rhino	31	2	2

Furthermore, the Spearman rank correlation coefficient is employed to examine the relationship among various ranking techniques. The resulting values are organized in Table 21. The findings indicate that the suggested technique demonstrates validity, showcases superior performance and secures the top rank.

Table 21
Comparative analysis using Spearman rank correlation.

Methods	Sum	Rank
WASPAS	3.6437	3
COPRAS	1.4833	5
WAS	3.3046	4
WAP	3.8711	2
Proposed	3.9001	1

Table 22
Comparison of proposed and ML-Predicted ranks for firefighting robots.

Robot	Proposed Rank	Predicted Rank
Thermite RS1	5	4.4763
Thermite RS3	1	1.4633
Colossus	8	6.9933
TAF60	7	6.7500
TAF60X	6	5.9933
TAF35	4	3.9367
Hercules	3	3.4367
Rhino	2	2.5733

5.5. Machine learning–based validation of the proposed decision-making model

A comprehensive model combining machine learning (ML)-based prediction and rank correlation analysis was employed to strengthen the validation of the proposed decision-making framework.

5.5.1. Validation protocol

To mitigate overfitting and ensure unbiased evaluation, a Leave-One-Out Cross-Validation (LOOCV) scheme was adopted. In each iteration, one alternative was excluded from the training set and used solely for testing to prevent any overlap between training and test data. A Random Forest Regressor was implemented to model the relationship between various MCDM scores (WASPAS, COPRAS, WAS, WAP) and the proposed IDFE–WISP rank. The hyperparameters such as the number of trees ($n_{\text{estimators}}$) = 300, maximum depth (max_depth) = 5, minimum samples to split (min_samples_split) = 2, minimum samples per leaf (min_samples_leaf) = 1, and random seed(random_state) = 42 were considered. The model learns the mapping from input features $X = \{x_1, x_2, x_3, x_4\}$, where x_i represents the score of each MCDM method and the output rank y produced by the proposed framework. The relation is defined by $\hat{y} = \mathcal{F}(X)$. The comparison between the proposed ranks and ML-predicted ranks is summarized in Table 22 and visualized in Fig. 9.

A strong positive correlation was obtained between predicted and actual ranks as the Spearman rank correlation coefficient was $\rho = 0.94$. All predicted ranks deviated by less than one position from the original, demonstrating high consistency.

Kendall's τ was computed between the proposed method and various MCDM approaches to provide a more rigorous comparative study. The results are shown in Table 23.

From the Table 23, it can be observed that the proposed method has a strong correlation with WAP ($\tau = 0.929$) while moderate correlation with WASPAS ($\tau = 0.643$) and WAS ($\tau = 0.500$) and lower correlation with COPRAS ($\tau = 0.429$). The absolute rank deviation calculated between the proposed method and various MCDM approaches is given in Table 24.

From Table 24, it is evident that Thermite RS3 exhibits perfect or near-perfect agreement across multiple methods, confirming it as a robust top-ranked alternative. wheres Colossus shows the largest deviation, indicating high sensitivity to methodological differences. Most alternatives have rank deviations ≤ 2, suggesting overall ranking stability.

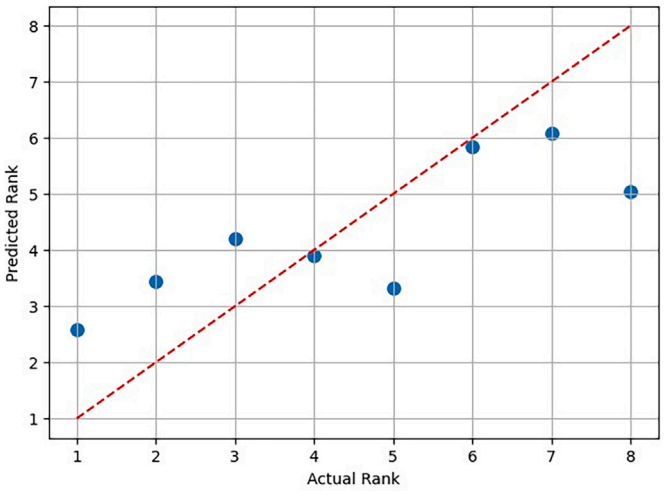


Fig. 9. LOOCV: Actual vs. Predicted.

Table 23
Spearman rank correlation matrix among MCDM methods.

Method	WASPAS	COPRAS	WAS	WAP	Proposed
WASPAS	1.000	0.071	0.857	0.714	0.643
COPRAS	0.071	1.000	−0.071	0.357	0.429
WAS	0.857	−0.071	1.000	0.571	0.500
WAP	0.714	0.357	0.571	1.000	0.929
Proposed	0.643	0.429	0.500	0.929	1.000

Table 24
Comparison of robot evaluations across MCDM methods.

Robot	WASPAS	COPRAS	WAS	WAP
Thermite RS1	3	2	3	0
Thermite RS3	0	2	0	0
Colossus	2	0	4	1
TAF60	1	1	1	1
TAF60X	1	4	1	0
TAF35	1	3	2	0
Hercules	1	2	2	0
Rhino	1	2	1	0

5.6. Results and discussion

This article examines strategies for reducing the death rate among people who work in hazardous fields like firefighting by integrating humans with robots to enhance human capabilities and improve their safety. Utilizing robots in a demanding firefighting scenario poses a complex challenge. Thus, the initial phase entails identifying the shortest path using the MO-IDF-BFS algorithm which integrates BFS with the MOIDFOM to refine risk and distance factors. Subsequently, the IDFE-WISP technique was utilized to prioritize the robots, facilitating the selection of the most suitable one among them. Considering the network G with 24 vertices and 31 edges as given in Fig. 4 the shortest path of the network is calculated using the BFS algorithm, where the source nodes H_1, H_2 and H_3 represent three hydrants to which the robots are connected and the sink nodes E_1 and E_2 represent the two entrances of the firing room. Thus, the paths from H_1, H_2 and H_3 to E_1 and E_2 were optimized and the values of the nodes a_i and a_j are assigned as $a_{ij} = \{\tilde{d}_{ij}, \tilde{r}_{ij}\}$, where $\tilde{d}_{ij}$ is the distance of the nodes and $\tilde{r}_{ij}$ is the risk level of the nodes. To value the path, the MO-IDF-BFS algorithm contributes a way to aggregate both the distance and risk level of the algorithm by utilizing MOIDFOM and the path $H_2 \rightarrow 17 \rightarrow 14 \rightarrow 12 \rightarrow 13 \rightarrow E_1$ is considered to be the shortest path with comparatively less risk. After identifying the shortest path, the next step involves selecting appropriate robots

to navigate the optimized route. The robots are chosen from a list and prioritized to assist firefighters in extinguishing the fire from a safe distance. Therefore, the IDFE-WISP model is employed for prioritizing the alternatives and in the robot selection process, Thermite RS3 is ranked as the top choice.

Additionally, a comparison study is carried out by comparing the proposed model with WASPAS, COPRAS, WAS, and WAP methods. In comparing the results, methods like WASPAS, COPRAS, WAS and WAP and the proposed model chose Thermite RS3 as their first priority. Since other robots vary with the ranking, the voting technique known as the Borda method is calculated to find the overall perfect ranking and was compared with the proposed method. From the results, it is observed that the first ranking remains the same. Furthermore, the Spearman rank correlation coefficient is employed to examine the relationship among various ranking techniques and the results indicate that the suggested technique demonstrates validity, showcasing superior performance and securing the top rank. Thus, the Thermite RS3 is found to be prioritized first among the others and the proposed algorithm is determined to be a better solving method for robot selection for the obtained shortest path.

Theoretically, this paper proposes new extended optimization and MCDM techniques under IDFS. Here, the MO-IDF-BFS algorithm is applied to find the shortest path and the proposed IDF-MCDM model is used, in which the criteria weights are obtained using the IDFE measure. The IDF-WISP model is then examined with new weights. With the help of the new score and accuracy ranking, the intuitionistic dense fuzzy utility values are combined and ranked. The combined ML-based validation and rank correlation analysis demonstrate that the proposed IDFE–WISP framework produces robust and internally coherent rankings, even with a small number of alternatives. The LOOCV scheme ensures out-of-sample evaluation, reducing overfitting and leakage risk. Overall, the proposed method effectively captures the decision pattern and maintains interpretability in firefighting robot selection.

6. Conclusion

This present study contributes to the advancement of Intuitionistic Dense Fuzzy Sets (IDFS) by developing an integrated decision-making framework. Three Intuitionistic Dense Fuzzy Entropy (IDFE) measures were formulated using max–min operations, exponential and logarithmic functions to effectively capture uncertainty within the IDFS environment. These measures enhance the ability to represent and discriminate complex decision information under uncertainty. To address multi-criteria path optimization problems involving both distance and risk, a Multi-Objective Intuitionistic Dense Fuzzy Breadth-First Search (MO-IDF-BFS) algorithm was proposed. The integration of this algorithm with the IDFE–WISP model resulted in a comprehensive AI-based decision-support system. Its application to a firefighting robot selection problem demonstrated the practical relevance of the approach, where the optimal path $H_2 \rightarrow 17 \rightarrow 14 \rightarrow 12 \rightarrow 13 \rightarrow E_1$ was identified and Thermite RS3 ranked as the most suitable alternative. Comparative analysis with the existing MCDM methods such as ARAS, WASPAS, COPRAS, WAS and WAP and the proposed method was conducted to validate the proposed method. Further validation was carried out by applying random forest regression to confirm the predictive reliability of the obtained rankings. Moreover, sensitivity analysis, rank correlation measures and Monte Carlo simulations provided additional evidence of the robustness of the model under varying conditions. These findings highlight the capability of the proposed model to support reliable and risk-aware decision-making in complex environments.

Despite its promising outcomes, this study recognizes several limitations. The entropy-based computations require notable processing time and a generalized framework for constructing new IDFE measures remains an open research direction. Moreover, while the BFS-based optimization approach performed effectively in this application, it may not scale efficiently for highly dynamic or large-scale decision environments.

Future work can focus on improving computational efficiency and scalability by incorporating advanced optimization techniques such as evolutionary or swarm-based algorithms. Further research may also explore the development of generalized entropy formulations and alternative similarity measures within the IDFS framework. Extending the applicability of the proposed methodology to broader real-world domains including autonomous navigation, disaster response and intelligent robotic systems would further strengthen its practical impact.

CRediT authorship contribution statement

Swethaa Sampathkumar: Writing – original draft, Visualization, Validation, Methodology, Data curation, Conceptualization. **Felix Augustin:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Data curation, Conceptualization. **Karthik Subramani:** Writing – original draft, Visualization, Validation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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